

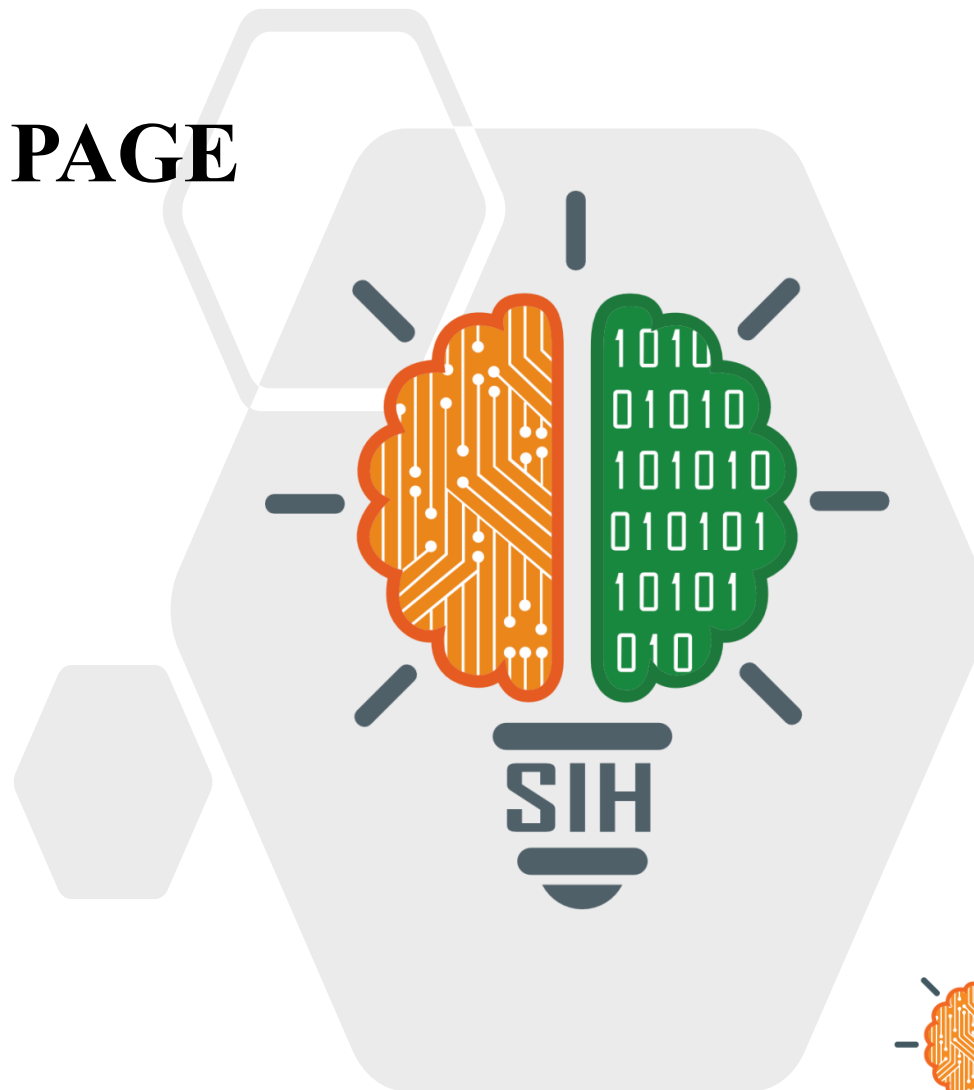


TITLE PAGE

SatQuery AI

Natural language in. Validated remote-sensing workflows out.

- Problem Statement ID – SIH26167
- Problem Statement Title- [FILL FROM SIH PORTAL]
- Theme- [FILL]
- PS Category- Software
- Team ID- [FILL]
- Team Name (Registered on portal) - [FILL]



SMART INDIA
HACKATHON
2026

Your
Team
Name

SATQUERY AI



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution
- How it addresses the problem
- Innovation and uniqueness of the solution

The model decides WHAT should happen · deterministic code verifies WHETHER it can safely happen

“Identify areas where vegetation decreased significantly between 2023 and 2025.”



Answer: loss percentage · hotspot map · per-region statistics · explanation · full provenance

INNOVATION AND UNIQUENESS

Reasoning ≠ verification

Model proposes; registry and validator decide. A reasoning error becomes a precise rejection, never a wrong map.

Capability-aware selection

Tools matched on declared contracts — modality, bands, image count, temporal pairing — not keyword similarity.

Composable operations

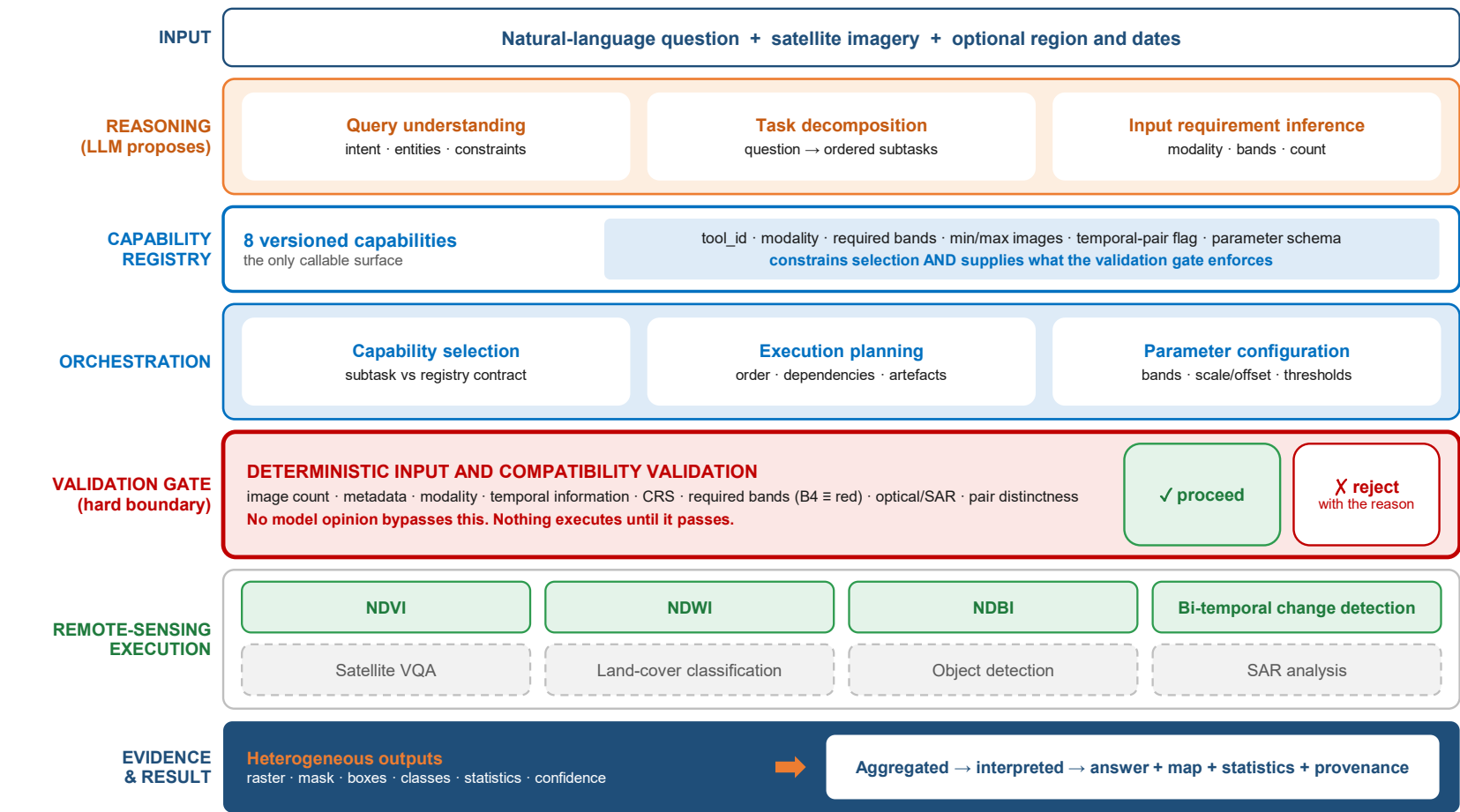
One question chains index computation, co-registration, differencing, thresholding and aggregation.

Extensible by declaration

A new model joins via a registry entry and a binding. Controller, validator and API unchanged.

- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

AI REASONING PROPOSES → DETERMINISTIC LAYERS VERIFY, EXECUTE AND EXPLAIN



TECHNOLOGIES

- Python 3 · Pydantic v2
typed schemas end to end
- FastAPI · Uvicorn
8-endpoint service layer
- NumPy
vectorised index arithmetic
- rasterio / GDAL
compressed GeoTIFF, JPEG2000, CRS
- Pure-Python GeoTIFF codec
dependency-free fallback backend
- LLM function calling
schema-constrained, registry-bounded
- pytest
326 automated tests

WORKING PROTOTYPE



- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

FEASIBILITY — ALREADY BUILT AND PROVEN

326

tests passing

8

capabilities declared

4

tools executing today

2

raster backends

VALIDATED

Ten-stage controller pipeline

understanding → planning → selection → validation → execution → integration

VALIDATED

Deterministic validator, 8 checks

count, metadata, modality, temporal, CRS, bands, optical/SAR, pairing

VALIDATED

Registry + implementation binding

no implementation can expose an undeclared capability

VALIDATED

NDVI, NDWI, NDBI on real rasters

sensor scale/offset, SCL and QA_PIXEL cloud masking

VALIDATED

Bi-temporal change detection

co-registration guard, Δ -index, σ -based thresholding

VALIDATED

Dual raster backend + HTTP service

tests prove both backends agree; 8 API endpoints

INTEGRATING

LLM reasoning and decomposition

pipeline already shaped for it; registry and validator keep the veto

NEXT STAGE

VQA, land-cover, detection, SAR

declared in the registry; executor reports them honestly as unbound

CHALLENGES, AND THE STRATEGY FOR EACH

Reasoning proposes an invalid workflow

The registry bounds what can be called; the validator bounds what can run. A bad plan is refused, not executed.

Heavy GDAL dependency blocks deployment

A dependency-free GeoTIFF path ships alongside rasterio, and tests prove the two backends agree.

Misaligned pair → confident, wrong change map

Grid, CRS, pixel size and origin are checked before arithmetic — and refuse rather than silently resampling.

Cloud contamination distorts index statistics

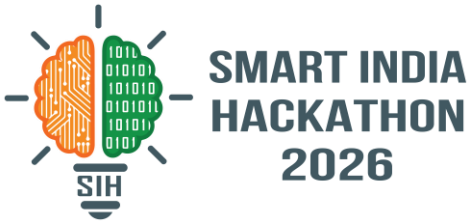
Sentinel-2 SCL and Landsat QA_PIXEL masking, plus baseline 04.00 scale/offset correction, before aggregation.

New models force a core rewrite

Adding a capability is a registry entry plus a binding. Controller, validator and API untouched.

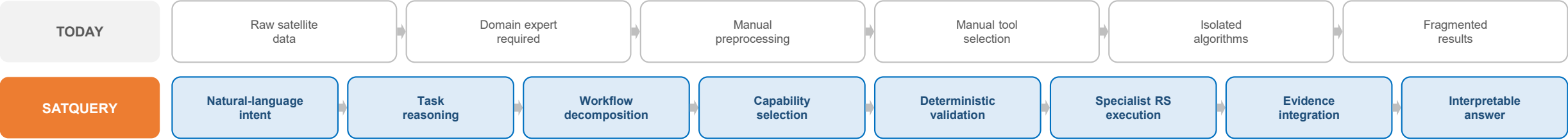
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IMPACT AND BENEFITS



- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

THE CHANGE SATQUERY MAKES



POTENTIAL IMPACT ON THE TARGET AUDIENCE



BENEFITS

Social

Geospatial evidence stops being the preserve of specialists. A district officer asks in their own words and gets an explained, defensible answer.

Economic

No repeated manual preprocessing or per-task scripting. One controlled workflow layer serves many departments instead of each rebuilding its own chain.

Environmental

Routine monitoring of vegetation loss, water extent and encroachment becomes cheap enough to run often — which is what turns monitoring into intervention.

Operational

Every answer carries its provenance: which tool ran, on which imagery, with which parameters. Reproducible and auditable, not one-off outputs.

• Details / Links of the reference and research work

WHY THE ARCHITECTURE IS SHAPED THIS WAY

Agentic planning and task decomposition

One question becomes an ordered set of executable subtasks instead of a single opaque call.

→ Reasoning layer

Schema-constrained tool calling

A model calling arbitrary functions is unsafe on real data. Calls are restricted to declared signatures.

→ Capability registry

Capability-aware orchestration

Selection follows each tool's declared input contract, not name similarity to the query.

→ Selection stage

Deterministic verification of model output

Models are strong proposers and unreliable verifiers, so admissibility is decided by code.

→ Validation gate

Remote-sensing foundation models and VQA

Pretrained geospatial and multimodal models supply scene-level understanding indices cannot express.

→ Execution layer (next stage)

Multi-temporal analysis and provenance

Change is meaningful only on co-registered pairs; an answer is trustworthy only when its path is recorded.

→ Execution + evidence integration

REFERENCES AND LINKS

• Yao et al., ReAct: Synergizing Reasoning and Acting in Language Models, ICLR 2023.	• Lobry et al., RSVQA: Visual Question Answering for Remote Sensing Data, IEEE TGRS 2020.
• Schick et al., Toolformer: Language Models Can Teach Themselves to Use Tools, NeurIPS 2023.	• Bastani et al., SatlasPretrain: Remote Sensing Image Understanding, ICCV 2023.
• Patil et al., Gorilla: Large Language Model Connected with Massive APIs, 2023.	• ESA, Sentinel-2 Level-2A Product Specification — Scene Classification (SCL), baseline 04.00.
• Jakubik et al., Foundation Models for Generalist Geospatial AI (Prithvi, IBM–NASA), 2023.	• USGS, Landsat Collection 2 Level-2 — QA_PIXEL bit definitions.



[QR]

GITHUB
github.com/
VinayakGupta-1/
SatQuery-AI

LIVE DEMO
[INSERT WEB DEMO LINK]